Conflict and Peace

Week 12: State capacity – Theoretical framework and weak states

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Drastic differences in capabilities of states across different countries

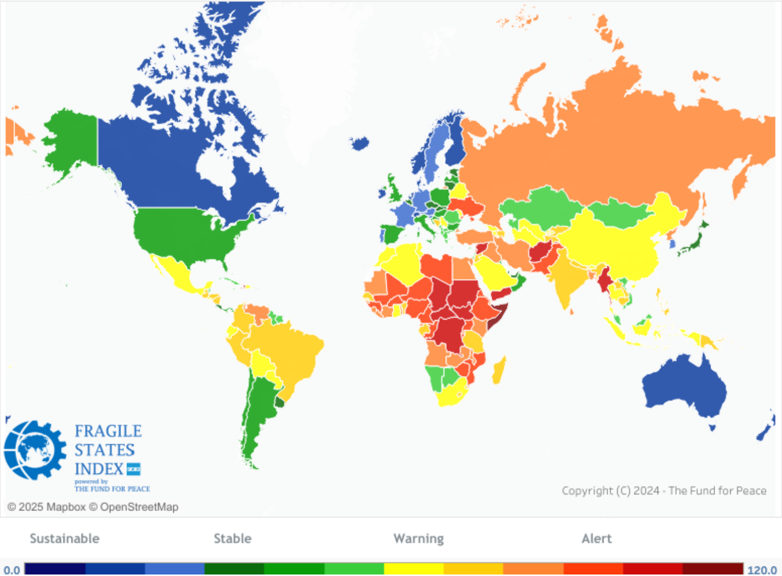
Some of the poorest countries in the world are also 'weak states'

- State capacity: Encompasses ability to support markets, provide public goods, and raise taxes from broad bases
  - Support markets: Enforce contracts, protect property rights
  - Provide public goods: Education, health, but also peace (by monopolizing violence)
  - Broad base taxes (income tax) are more rule-based, less burdensome for individuals, but require large investments for implementation
  - Narrow base taxes (royalties and tariffs) are easy to implement but could be corruptible and extracts particular group excessively
- Weak states: States that cannot support basic economic function, raise revenues, deliver essential public services, and provide law and order

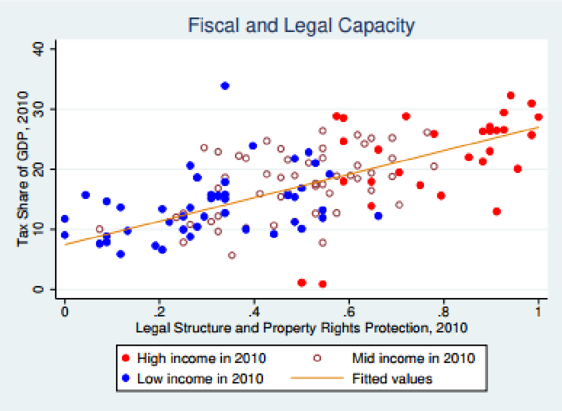
State capacity is widely dispersed, but also clustered

- Weak states are also poorer and mired in internal conflicts
- Developed countries: High income, functioning institutions, less internal conflicts

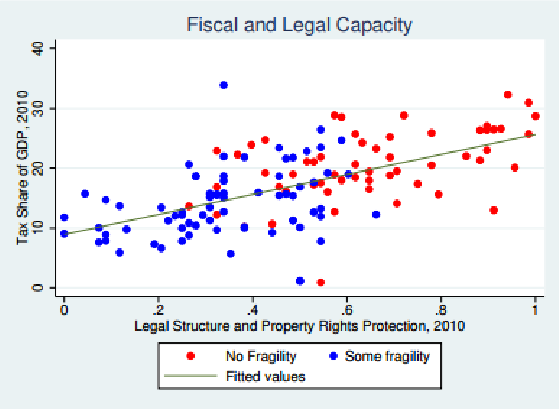
Huge variation in state capacity across the globe



State capacity clustered with income and functioning of institutions

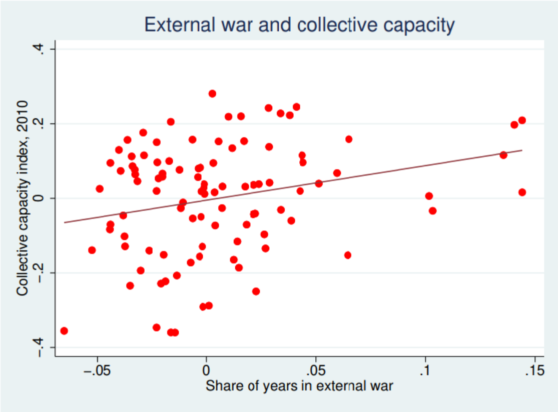


(a) Income and capacities are clustered

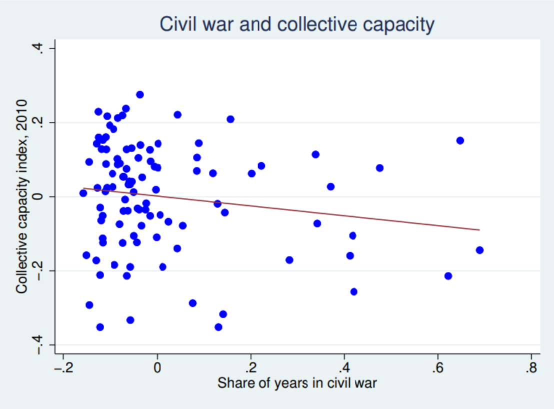


(b) Capacities and fragilities index

External wars increase capacity; internal wars hamper it



(a) External wars positively correlate



(b) Internal wars negatively correlate

Why care about state capacity, and open questions?

Why do state capacities matter?

- Capable states can distribute fruits of economic development and prevent violence that stifles the progress
  - Ability to tax, enforce law, and distribute services contributes to growth and peace
- Ultimately relates to economic and institutional development: Many weak states are poor, authoritarian, and in internal conflict

We discuss open questions on state capacity with theories and empirical findings

- What forces explain the formation of capable states?
- How do weak states perpetuate violence, corruption, and weak growth?
- What explains the clustering of state institutions, violence, and income?

Roadmap for today

1. Overview of the topic

2. Conceptualizing state capacity
   2.1 Besley, Persson (2009): The Origins of State Capacity: Property Rights, Taxation and Politics
   2.2 Gennaioli, Voth (2015): State Capacity and Military Conflict
   2.3 Besley (2020): State Capacity, Reciprocity, and Social Contract
   2.4 Rogowski, Gerring, Maguire, Cojocaru (2022): Public Infrastructure and Economic Development: Evidence from Postal Systems [student presentation]

3. Symptoms of Weak States
   3.1 Shleifer, Vishny (1993): Corruption
   3.2 Dal Bó, Dal Bó, Di Tella (2006): Plata o Plomo? Bribe and Punishments in a Theory of Political Influence
   3.3 Sanchez de la Sierra (2020): On the Origins of the State: Bandits and Taxation in Eastern Congo [Student Presentation]

Besley, Persson (2009):
The Origins of State Capacity: Property Rights, Taxation and Politics

Q: What determines the state's investment in fiscal and legal capacity

How are institutional capacity support markets and collect taxes developed?

- Historians note the development of capacity in response to external wars (Tilly 1990)
- Social sciences (economics, especially) typically assume such capacities exist by default
- In reality, institutional capacities differ starkly

This paper provides theoretical framework of state institutions as endogenous investments

- Individuals come from different groups (ruling vs nonruling, different wealth)
- State collects taxes, protects property rights, and provide public goods
- What determines taxation, public goods provision, and support for market activities
- Why are there differences in investments in these capacities across different settings?
  - Focus on high capacities of wealthy, democratic, and parliamentary democracies
- Provides grounds for studying how institutions evolve during development

Setting up the framework: Definitions and layout

What makes up state capacity

- Legal capacity: Enforce contracts, regulate markets, and protect property rights
- Fiscal capacity: Collect taxes to finance public goods and 'redistribution'

Setting the model up

- There are two time periods $s \in \{1, 2\}$ and two groups of individuals $J \in \{A, B\}$
- Individuals have linear preferences over public goods and post-tax income
- State (led by A or B) protects property rights, collect taxes in both periods
- State can also increase its legal and fiscal capacities at $s = 1$
- State maximizes (politically) weighted sum of individual utilities under budget and capacity constraints

Setting up the framework: Individual utilities

Individual utilities (For given group $J \in \{A, B\}$)

- Gain utility from public good G_s (by factor of α_s) and income Y_s^J (taxed by t_s^J)

$$v_s^J = \alpha_s G_s + (1 - t_s^J) Y_s^J (p_s^J)$$

- Income gain is higher with increase in p_s^J, enforcement of property rights, $\left(\frac{\partial Y_s^J}{\partial p_s^J} > 0\right)$
- Taxes deduct from income but used to finance public goods

Differences across groups

- Demographic share: β^J of population in group $J \in \{A, B\}$ ($\beta^A + \beta^B = 1$)
- Different protection, taxation (negative taxes possible), and income
- Public good is distributed commonly (no differences in level of public goods)
- Assume that group A is ruling at $s = 1$, with chance of continued ruling in $s = 2$ is γ_A

Setting up the framework: State objectives and constraints

Objective of the state: Politically weighted sum of individual utilities

- $\bar{\rho}$ and $\underline{\rho}$: Relative political weights on ruling and nonruling groups ($\bar{\rho}\beta_s^A + \underline{\rho}\beta_s^B = 1$)

$$V_s = \alpha_s G_s + \bar{\rho}\beta_s^A(1 - t_s^A)Y_s^A(p_s^A) + \underline{\rho}\beta_s^B(1 - t_s^B)Y_s^B(p_s^B)$$

  - $\bar{\rho}$ captures how A values its own group relative to demographic share (1 or greater)
  - $\underline{\rho}$ captures how A values nonruling groups (less or equal to 1)
- At $s = 1$, invest in legal (π_s) and fiscal capacity (τ_s) at cost of $L(\pi_2 - \pi_1)$ and $F(\tau_2 - \tau_1)$

Constraints of the state

- Capacity constraints: $p_s^J \leq \pi_s$, $t_s^J \leq \tau_s$
- Budget Constraints
  - $s = 1$: Capacity investment included $\rightarrow \sum_J \beta^J t_1^J Y_1^J = G_1 + L(\pi_2 - \pi_1) + F(\tau_2 - \tau_1)$
  - $s = 2$: No new investment$\rightarrow \sum_J \beta^J t_2^J Y_2^J = G_2$

Policy choices in utilitarian vs politically-motivated settings

Policy choices on p_s^J and t_s^J in utiltarian governments ($\bar{\rho} = \underline{\rho} = 1$)

- p_s^J increases indirect utilities: Provide to the fullest possible $p_s^J = \pi_s$
- t_s^J works in both ways - incomes are taxed, but used for public goods
- $\alpha_s \geq 1$: Tax to the fullest possible ($t_s^J = \tau_s$), all revenues to G_s and investments
- $\alpha_s < 1$: $G_s = 0$, smallest possible taxes (0 in $s = 2$, exact cost of investments in $s = 1$)

Politically-motivated objective maximization ($\bar{\rho} > 1 > \underline{\rho}$)

- No different logic in legal capacity: Higest protection possible
- Public goods are provided only if ruling class values it sufficiently ($\alpha_s \geq \bar{\rho}$)
- If $\alpha < \bar{\rho}$: $G_s = 0$, and taxes could be 'redistributed' from nonruling to ruling class
  - Ruling group gains $\bar{\rho}$ from 'redistribution', nonruling loses $\underline{\rho}$ from this
  - With $\bar{\rho} - \underline{\rho} > 0$ beneficial to extract (from nonruling group standpoint)
  - Implies that in this setting, people have different interests towards fiscal capacity

Investing in state capacity

Investments in fiscal (τ) and legal (π) capacities maximizes expected social utility

- Factors: Wealth (Y_s^J), Value of public goods (α_s), Regime stability (γ_A), representativeness ($\bar{\rho} - \underline{\rho}$)
- Legal capacity is desirable in any case
- When $\alpha_s < \bar{\rho}$, high τ undesirable for non-ruling group $\rightarrow$ Underinvest in capacities
  - In other cases, no conflict over the desirability of fiscal capacity

Key findings (in theory and empirics)

- In most cases, legal and fiscal capacities complement each other
  - $\pi \Uparrow \rightarrow$ More revenues for public goods $\rightarrow \tau \Uparrow \rightarrow$ More funds to increase protection $\rightarrow \pi \Uparrow$
- What factors increase investments in state capacities?
  - High Y_s^J, High α_s, High γ_A, Small $\bar{\rho} - \underline{\rho}$
  - Mismatch in political power and economic influence $\rightarrow$ Fiscal capacity underinvested

Theoretical prediction matches well with observed correlates

TABLE 1—ECONOMIC AND POLITICAL DETERMINANTS OF LEGAL CAPACITY

	Private credit to GDP (1)	Ease of access to credit (country rank) (2)	Investor protection (country rank) (3)	Index of government anti-diversion policies (4)
Incidence of external conflict up to 1975	0.510***	0.647**	0.029	0.576***
	(0.143)	(0.191)	(0.209)	(0.170)
Incidence of democracy up to 1975	0.953	0.110	−0.044	0.126**
	(0.059)	(0.267)	(0.078)	(0.050)
Incidence of parliamentary democracy up to 1975	0.001	0.145	0.339**	0.112*
	(0.063)	(0.114)	(0.137)	(0.061)
English legal origin	−0.009	0.068	0.125**	−0.007
	(0.033)	(0.057)	(0.063)	(0.040)
Socialist legal origin	—	0.098	0.097	0.010***
		(0.111)	(0.115)	(0.035)
German legal origin	0.406***	0.295***	−0.008	0.248***
	(0.120)	(0.064)	(0.149)	(0.053)
Scandinavian legal origin	0.112***	0.204***	0.087	0.254***
	(0.041)	(0.067)	(0.098)	(0.055)
Observations	93	122	120	115
R^2	0.524	0.334	0.256	0.596

Theoretical prediction matches well with observed correlates

Table 2—Economic and Political Determinants of Fiscal Capacity

	One minus share of trade taxes in total taxes (1)	One minus share of trade and indirect taxes in total taxes (2)	Share of income taxes in GDP (3)	Share of taxes in GDP (4)
Incidence of external conflict up to 1975	0.762***	0.598***	0.579***	0.555***
	(0.250)	(0.241)	(0.220)	(0.162)
Incidence of democracy up to 1975	0.143	−0.078	0.091	0.088
	(0.077)	(0.100)	(0.059)	(0.059)
Incidence of parliamentary democracy up to 1975	0.031	0.122	0.212***	0.160**
	(0.083)	(0.103)	(0.078)	(0.068)
English legal origin	−0.038	−0.012	−0.034	−0.015
	(0.058)	(0.061)	(0.043)	(0.042)
Socialist legal origin	0.136**	−0.222***	−0.109***	−0.119
	(0.058)	(0.037)	(0.065)	(0.031)
German legal origin	0.175***	0.196***	0.171*	0.010***
	(0.052)	(0.090)	(0.010)	(0.083)
Scandinavian legal origin	0.189**	0.068**	0.258**	0.292***
	(0.077)	(0.084)	(0.134)	(0.087)
Observations	103	103	103	103
R^2	0.356	0.305	0.600	0.576

Gennaioli, Voth (2015):
State Capacity and Military Conflict

Q: What explains the rise of centralized states in Europe?

Centralized states are relatively recent concept

- Before: No professional bureaucracy, limited tax powers, armies of mercenaries
- Europe began to form centralized fiscal appartus from 1500s
- Key driver: External wars - but really?
  - Many wars pre-1500s, and many states (Spain, Austria) late in developing fiscal capacity
  - War may not be for common interests (personal glory)
  - Military capacity $\rightarrow$ war (reverse causality feasible)

This paper develops and tests the model of different types of wars in state-building

- War differs by how much money matters for military success
- Cohesiveness of the pre-war state also important
- Tests this on data of 374 battles in Europe since 1500
- With rising importance of money in wars, cohesive states developed more

Historigal background and theoretical framework

European wars and fiscal capacity

- Role of money became important around 1500s (but not before)
  - Military Revolution: Standing army and new technology (gunpowder) made money crucial
  - Divergence of military powers followed (ENG, FRA vs SPA, AUT, POL)
- Some states increased taxation since 1500s, but large cross-country variations
  - 10-fold increase for England, Decrease for Ottoman Empire and Poland
  - Reliance on domain income $\rightarrow$ systemized tax collection (income, farming-based)

Theoretical model: Centralized vs decentralized state without external wars

- There are local, market, and home activities (not taxed)
- Centralized: Local and market activities taxed by central ruler
- Decentralized: Market activities taxed across different localities (Thus overtaxed)
  - Local production only taxed at base. Local powerholder collects taxes and gets to keep it
  - Cost of centralization: Need to fend off local powerholders who lose tax revenues

External wars and centralized state capacity

How does external war come into play?

- War absorbs revenues of warring rulers + Redistributes revenues from loser to winner
- Winning depends on military strength (determined by manpower and expenses)
- War threat affects centralization incentives in two ways
  - Increase centralization: Rising revenues makes it more likely to win
  - Decrease centralization: Revenues spent on war, and risk of larger losses after war
- Determinant of divergent patterns of state-building
  - Higher money sensitivity: Increase incentive to centralize
  - Cohesiveness: External war becomes common threat $\rightarrow$ more incentive to centralize
  - Countries less likely to win (divided to begin with) unlikely to centralize
- When money does not matter: Less incentive to centralize

Data and empirical strategy

Variables and data sources (Jacques (2007), Karaman, Pamuk (2010))

- Data on 374 European wars as external wars
- Fiscal capacity: Tax revenues per capita divided by wage (mimic share of taxes)
- Cohesion measured based on number of independent predecessor states, ethnic fractionalization measure, and surface area

Empirical strategy seeks to estimate correlates between money sensitivity, cohesion, and fiscal capacity

- Primarily use OLS (along with country fixed effects)

Empirical finding: Richer countries won more wars after 1650

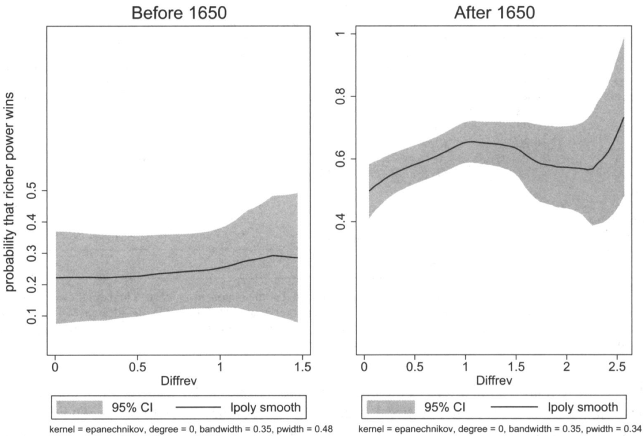


FIGURE 5

War and fiscal resources

Notes: The graph shows the smoothed local polynomial and 95% confidence intervals. The y-axis measures the probability of the richer power winning in battle. Note that battle outcomes are coded as either victory or draw, meaning that the probability of the poorer power winning is not equal to 1-probability of richer power winning. The x-axis shows the log difference in total fiscal revenue, calculated as Diffrev=$ln(T_{H,t}) - ln(T_{L,t})$, where $T_{H,t}$ is the tax revenue of the fiscally stronger power, $T_{L,t}$ is the revenue of the fiscally weaker power.

Sufficient cohesion and money sensitivity matters

TABLE 5

Revenue raising and the military value of money (dependent variable: tax pressure revenue per capita)

	(1) OLS	(2) FE	(3) FE+institutions	(4) Controls
Panel A				
λ_t	66.45**	59.66**	57.42**	55.47**
	(2.85)	(2.5)	(2.5)	(2.75)
ConsExec			1.18**	
			(2.89)	
Area				$-1.21e-12$
				(-1.35)
Slope				-0.137
				(-0.54)
Pop_{200}				20.64***
				(4.22)
Constant	-2.9	-2.1	-2.9	-4.8
	(0.81)	(0.59)	(0.8)	(1.5)
N	53	53	53	53
R^2	0.11	0.74	0.76	0.62

	(1) Low-fraction Sum^H=0	(2) High-fraction Sum^H=1	(3) Inter	(4) Inter+FE
Panel B				
λ_t	75.8*	30.85*	82.46**	75.77*
	(2.4)	(2.2)	(2.67)	(2.4)
Sum^H			2.5	
			(0.6)	
λ*Sum^H			-56.6*	-44.92
			(-2.3)	(-1.9)
Constant	-4.8	1.97	-3.16	-4.82
	(-0.97)	(0.9)	(-0.64)	(-0.96)
N	34	19	53	53
R^2	0.64	0.77	0.39	0.75

Notes: * $p<0.1$, ** $p<0.05$, *** $p<0.01$; t-statistics in parentheses; λ is the estimated slope parameter from a regression of military outcomes on relative fiscal strength, *ConsExec* is constraints on the executive, *Area* is the total surface area of a state, *Slope* is the average slope of terrain (a measure of ruggedness), Pop_{200} is the share of the population within 200 km of the capital, and Sum^H *is the sum of standardized measures of fractionalization.*

Takeaway and discussion

Not all external wars affect state capacity equally

- Early history wars did not contribute to fiscal capacity
- When revenues mattered in winning wars, it increased incentive to centralize
- The effect is augmented with initial cohesion
  - These patterns explain why centralization patterns diverged even in Europe

Remaining questions

- With decreased frequencies of war, what explains development of fiscal capacity?
  - Ability to implement broad-based tax still key
  - Potential role for digitalized tax system (Pomeranz 2015; Okunogbe, Tourek 2024)
- Maybe investment in military capacity is a result of possible external war
  - If military investment = state capacity, then Besley, Persson (2009) causal logic (war $\rightarrow$ invest capacity) still applicable

Besley (2020):
State Capacity, Reciprocity, and Social Contract

Q: How does civic culture contribute to fiscal capacity?

Increased tax intake enabled governments to spend on essential services

- Not just on national defense, but also health care, education, income transfers etc
- Switches in public spending requires economic, political, and social changes
  - Should develop capacity to tax and raise revenues
  - But also, cultivate civic culture that increases tax *compliance*

This paper extends the Besley, Persson (2009) model

- By incorporating the role of civic culture, reciprocity, and social contract
- Relative payoffs of civic-minded citizens and materialistic citizens matter
- Investigates how civic culture can develop in the long-run
- Ultimately answers how state capacity can persist with interactions with citizens

Motivating findings: Different attitudes to tax compliance

DETERMINANTS OF ATTITUDES TOWARDS TAX COMPLIANCE[a]

Low confidence in government	–	−0.177(0.029)	–
Low trust in people	–	–	0.009 (0.022)
Male	−0.212 (0.018)	−0.210 (0.018)	−0.212 (0.018)
Age 30–49	0.222 (0.021)	0.220 (0.021)	0.222 (0.021)
Age 50+	0.571 (0.043)	0.564 (0.043)	0.572 (0.043)
Education: Middle	0.042 (0.026)	0.046 (0.026)	0.042 (0.026)
Education: Upper	0.097 (0.032)	0.101 (0.033)	0.098 (0.031)
Income 2	−0.017 (0.055)	−0.014 (0.029)	−0.017 (0.055)
Income 3	−0.034 (0.054)	−0.032 (0.054)	−0.034 (0.053)
Income 4	−0.087 (0.056)	−0.087 (0.057)	−0.087 (0.056)
Income 5	−0.043 (0.056)	−0.041 (0.057)	−0.043 (0.056)
Income 6	−0.141 (0.056)	−0.141 (0.057)	−0.141 (0.056)
Income 7	−0.161 (0.057)	−0.160 (0.057)	−0.161 (0.057)
Income 8	−0.158 (0.069)	−0.157 (0.069)	−0.158 (0.069)
Income 9	−0.194 (0.063)	−0.191 (0.062)	−0.193 (0.063)
Income 10	−0.234 (0.081)	−0.236 (0.080)	−0.233 (0.081)
Number of observations	303,229	303,229	303,229

[a]The dependent variable is based on question "Is it justifiable to cheat on your taxes if you have a chance?" from the World Values Survey and European Values Survey (various waves) with the scale reversed so that the highest score is associated with cheating not being justified. All specifications include wave and country dummies with standard errors clustered at the country level. The data cover

Source: Individual responses from World Value Survey, regressing compliance on individual characteristics, country and wave dummies. Income is displayed from lowest to highest

Model setup

Basic structure of the model

- Share e of elites who decides public goods and transfers, and non-elites
- Among non-elites, μ_s of civic-minded citizens and $1-\mu_s$ of materialists
  - All subject to taxes, with possible noncompliance (n), and receive utility from public goods
  - Civic-minded: Gains if tax used on public goods (G), loss if used on transfers to elites (B)
  - Noncompliance: Incurs cost $C(n)$, which rises with fiscal capacity τ (easily detected)
  - Individual Utility: $\lambda > 0$ for civic-minded citizens, $\lambda = 0$ for materialists

$$\alpha G + b + w(1 - (1-n)[t - \lambda\{G - eB\} - \tau C(n)])$$

- Revenues must be spent on public goods G, transfers to elites (B) and non-elites (b)
  - For every B, share $0 \leq \sigma \leq 1$ must be spent on b. Thus $b = \sigma B$ (Higher $\sigma \rightarrow$ cohesiveness)
  - Budget: $eB + (1-e)\sigma B = T - G$
- Fiscal capacity is determined by physical investments but also 'tax morale'
  - For materialists, only physical investments matter
  - For civic-minded person, public good$\uparrow \rightarrow$ tax morale $\uparrow$ (transfer$\uparrow \rightarrow$ tax morale $\downarrow$)

Optimal policy choice

Optimal policy choice

- Elites choose t, G, and B to maximize the sum of individual utilities
- Subject to cohesiveness of institutions (σ), fiscal capacity (τ) and civic culture (μ)
- Whether revenues are spent on G or B depends on cohesiveness and civic culture
  - Public goods are more likely to be provided in scenario where civic culture ($\mu \Uparrow$) and cohesiveness ($\sigma \Uparrow$) are dominant (and also raise fiscal capacity)
  - This comes from the fact that civic-minded persons are more likely to comply if revenues are used for public goods (and vice versa)
  - Furthermore, high cohesiveness makes transfers to elites expensive $\rightarrow$ make public goods relatively cheaper
- Over time, additional public goods due to $(\mu, \sigma) \Uparrow \Rightarrow$ benefits of being civic-minded $\Uparrow$
  - Conversely, narrow range of public goods provision from $(\mu, \sigma) \Downarrow \Rightarrow$ civic-mindedness $\Downarrow$
- Initial state of civicness, cohesiveness + strength of common interest (α) matters

Core implications of the model

Cohesiveness of institutions and civic culture is complementary

- Cohesiveness of institutions are strengthened when more civic-minded persons induce high public goods over transfers to elites
- People are more likely to be civic minded if institutional setups make transfers to elites costly (through cohesiveness)

These determine how common interests develop

- In a less cohesive, weakly civic-minded setting, external threat may not significantly increase common interest

Civic-mindedness and cohesivess can be fostered in more representative political setup

- Elites that more closely represents non-elites more likely to sustain power
- Further increases returns for being civic-minded

Potential extensions

High dependence on outside sources of funds

- Outside sources of funding reduces need for own-source tax revenues
- This reduces importance of tax compliance and the role of civic culture
- Sudden, exogenous changes to outside funds $\rightarrow$ Challenges in raising fiscal capacity

Can elites determine civic-mindedness (μ)?

- We assumed that μ is exogenously determined
- But these can be potentially determined by elites
- Room for extension: Making tax avoidance costly, Benefits of public services over time

Takeaways: How these topics can be empirically studied

How can fiscal capacities be built

- Evolution towards broad-based system (Gordon, Li 2009; Jensen 2022)
- Efforts to gather more accurate information in developing countries (Balan et al 2021)

What increases tax morale and compliance?

- Increase benefits of compliance (tax breaks for formal firms) - (Benhassine 2018)
- Tax perceptions and electoral outcomes (Christensen, Garfias 2021)

Other factors that contribute to state capacities

- Role of bureaucrats: Well-motivated, well-trained bureaucrats matter for effective public goods provision
- Active studies on how to recruit and monitor them (Dal Bó et al 2013)

Rogowski, Gerring, Maguire, Cojocaru (2022): Public Infrastructure and Economic Development: Evidence from Postal Systems

Shleifer, Vishny (1993): Corruption

Q: What conditions increase possibility of corruption

Definition and setup on corruption

- Corruption: Sale by government officials of government property for personal gain
    - Collecting bribes for providing permits, custom passage, bar competitor entry
    - Defense officials selling contracts for personal gain
- In some cases, bribes *may* account for a large fraction of national income
    - In reality, really difficult to track. Most estimates may be underestimated

This paper provides framework on the organization and effects of corruption

- Compare different types of bribes (theft by officials vs not)
- Explain the distortionary effects of bribes
    - Bribe “greases the oil?": Overcome cumbersome regulation etc
    - Slows down development, imposes additional costs

Baseline model

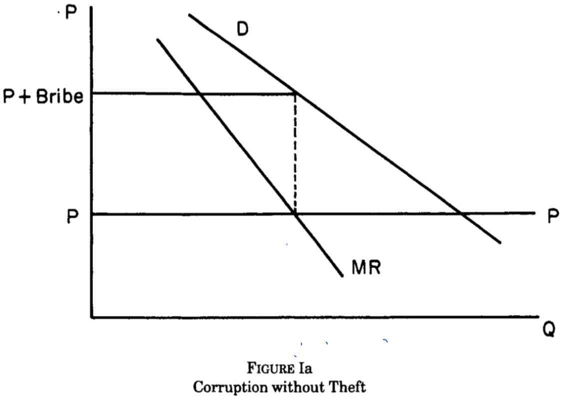
Model setup

- Government good is up for sale (say, a passport or a license)
- Official in charge can deny the sale of this good without risk of detection by its boss
- Consider the official as monopolist: Maximize values of bribe for selling the good

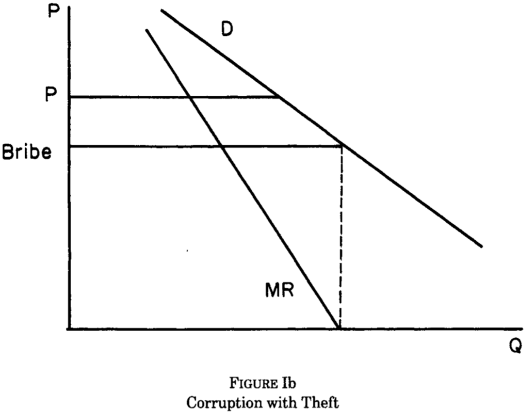
Cost-benefit problem of the official

- Public good has a demand function $D(p)$, where p is the official government price
- For the official, no cost in *producing* the good (already made by government)
- Marginal cost of selling the good is different case by case
  - Official turn over p to government per sale (no theft): p per sale $\rightarrow$ Buyer price $> p$
  - Officials turn nothing to government (theft): 0 marginal cost $\rightarrow$ Buyer price $< p$
- Create a shortage of goods (first case) & Reduce observance of law (second case)

Buyer price when corruption is present



(a) Without theft



(b) With theft

What if you need multiple bribes to get a good?

Suppose you need more than one licenses

- Two goods (x_1, x_2) are involved, with bribe and official prices of (p_1, p_2) and (MC_1, MC_2)
- Officials may act independently or collude
  - In former, each sets their own bribe $\rightarrow$ bribe prices are higher $\rightarrow$ underprovision of goods
  - In latter, keep bribe on one good down and profit off from bribe on the other
- If a given good is supplied by different competing officials, level of bribes $\Downarrow$
  - If someone asks for a bribe, go to the other official that do not ask for observance
- The first two cases lead to underprovision and higher bribes compared to the third

What determines which one of the above structure are realized?

- How responsive is public pressure against corruption? (e.g active monitoring)
- How active is the political competition (which leads to pressure against corruption)

How does corruption further distort outcomes?

Corruption is usually done in secret

- Thus, more actions need to be carried out by officials to avoid detection
- May involve inducing substitutions into goods where bribes are hard to detection
  - Results in higher bribes (to cover costs) and further underprovision of certain goods
- These actions increase social costs, much more than taxes

Other costs incurred due to secrecy

- Reluctance to change and innovation (keep outsiders off)
- Reluctance to being open (hinder efforts for transparency)

Updated analysis on corruption so far

How to curb incentives for corruption among bureaucrats?

- Using data from public procurement (Best et al 2023)
  - Breaks down prices charged in procurements to corruption and non-corrupt factors
- Using bureaucrat placement (Khan et al 2019; Xu et al 2021)
  - Finds that those assigned to home states are less effective
  - Other evidence that performance-based postings lead to better outcomes

How to increase monitoring against corruption?

- Effects of media reports on capture of public funds (Reinikka, Svensson 2005)
  - Provide information on local officials' usage of funds $\rightarrow$ reduce wasted expenditures
- Effects of reports on corrupt officials (Ferraz, Finnan 2008; Larreguy et al 2020)
  - Publicize audit on local officials' performance $\rightarrow$ Accountability $\Uparrow$ ($\because$ electoral outcomes)

Dal Bó, Dal Bó, Di Tella (2006): Plata o Plomo? Bribe and Punishments in a Theory of Political Influence

Q: How do state capture affect quality of public officials

In weak states, politicians are subject to punishments by private entities and bribes

- Many private entities seek to influence politicians through bribes and coercion
- Punishments and bribes complement: Threat of punishment makes bribing easier
- Yet, there are things where models with only corruption cannot explain
  - Why give officials more discretion and immunity?
  - When is it justified for the state to monopolize violence?

This paper updates framework of political influence to include bribes and coercion

- Investigates how possibility of coercion affects quality of public workers
- Explains why political immunities exist in contexts where private coercion is prevalent
- Explains how discretion actually makes public work less attractive
  - In model with bribes only, immunities are rarely justified (weaken monitoring mechanisms)
  - Also in the old model, discretion increases corruption and appeal of public work

Model setup: Decision to join public work

First stage: Citizens decide to enter public office or not

- Citizens endowed with ability $a \in [0, \infty)$ (a probability distribution)
- Alternative is to choose private sector (which gives wage a, equivalent to their ability)
- Return from public work: public sector wage (w) + behavior of pressure group
- Citizens with $a \leq w$ apply (and w types selected, assuming recruiters observe types)

Second stage: Pressure group has an opportunity to influence officials

- Public workers provide public good, whose quality positively correlates with ability
- Have discretion to allocate (π) towards pressure group, with cost c if caught
- With probability γ, officials decide to accept bribes or face punishments
  - $1 - \gamma$: Only punishment (no possibility to accept bribes) - when bribes are infeasible
- Pressure groups can either bribe (b) or punish (p) at costs $\beta\Phi(b)$ and $\rho\Psi(p)$

Analysis of payoffs

Decision on accepting (and handing out) bribes

- Officials facing pressure groups accept bribes (share γ) if

$$w + b - c \geq w - p$$

- Pressure groups sets b and p to maximize the following profits

$$\Pi(b,p) = \gamma(\pi - \beta\Phi(b)) - (1-\gamma)\rho\Psi(p) \ \ \textit{subject to } b \geq c - p$$

- $\tilde{\pi}$: Minimal discretion required for pressure groups to influence officials (break-even)
  - Inversely related to state capture: $\tilde{\pi} \Uparrow \rightarrow$ Cost of bribing an official $\Uparrow \rightarrow$ State capture $\Downarrow$

If only bribes are available ($p = 0$ for all cases)

- Solve $\gamma(\pi - \beta\Psi(b))$ subject to $b \geq c$ $\rightarrow$Bribe just enough to cover c, ($b = c$)
- Official payoffs/skill is w, and $\tilde{\pi}^0 = \beta\Phi(c)$
- Drop in bribe costs ($\beta \downarrow$) $\rightarrow$ Higher state capture ($\tilde{\pi} \Downarrow$), no effect on quality of officials

What happens when both bribes and coercion are possible?

Problem when both b and p are active

- Pay bribes just enough to cover $b = c - p$ (Bribes become cheaper)
- The shortfall is made up with nonzero punishment p^*, such that $b + p^* = c$
- $\tilde{\pi}^p$ satisfies $\frac{1-\gamma}{\gamma}\rho\Psi(p^*) + \beta\Phi(c - p^*) \rightarrow$ Any discretion π larger than this, then influence
- Resulting equilibrium payoff of official if pressure groups are active
  - If official refuses bribe, payoff is $w - p^*$ (lower due to nonzero punishments)
  - If bribes are accepted, payoff is $w + b - c = w + c - p^* - c = w - p^*$
- When pressure groups are inactive: payoff is $w > w - p^*$
- Thus, the quality of bureaucrats fall in this scenario
- Furthermore, $\tilde{\pi}^p < \tilde{\pi}^0 \rightarrow$ When threats are usable, more state capture!

Negative effects amplified with cheaper ways to influence officials

Exogenous changes in cost of influence (Suppose that β and ρ drops)

- Recall that $\tilde{\pi}^p = \frac{1-\gamma}{\gamma}\rho\Psi(p^*) + \beta\Phi(c - p^*)$ and $\Pi(b,p) = \gamma(\pi - \beta\Phi(b)) + (1-\gamma)\rho\Psi(p)$
- Fall in any one of β and ρ raises profits and decreases $\tilde{\pi}^p$
  - The latter implies state capture will be more rampant
  - Existing pressure groups ramp up activity + new pressure groups enter
- If primarily driven by $\rho \Downarrow$, then payoff of officials (and quality) definitely falls
  - Usage of coercion increases in any case
- If primarily driven by $\beta \Downarrow$, change to pay off ambiguous (but definitely not increase)
  - Existing groups shift away from coercion $\rightarrow$ Minimal effect on wages
  - New entrants that use both coercion and bribes $\rightarrow$ lower payoffs

Implication on discretions

How do discretion affect quality of politicians

- If discretion $\pi < \tilde{\pi}^p$, less incentives for pressure groups to influence officials
- Higher discretion could lead to more waste (more corrupt decision)
- As more groups with coercion decrease payoffs, quality of officials worsened

Immunity of officials (in principle)

- Can insulate honest officials from outside pressures, mitigate false accusations
- But can mask corrupt officials
- In setting with rampant coercion from pressure groups
  - There are greater effect of protection to officials with immunity
  - This can overweigh bad effects of masking corrupt officials
- When legal system is effective, masking effects outweigh pressure effects

Takeaways and discussions

Threat of coercion imposes negative effects on governance

- Reduces payoff for officials $\rightarrow$ lower quality of officials
- More influence from private pressure groups $\rightarrow$ Increased state capture
- Augments effects of bribes: Cheaper to bribe officials

Remaining questions

- Ultimately, need to improve legal capacity
  - To prevent pressure groups and detect corruption
  - But transitioning to better legal system is tough
- Incentives of bureaucrats: How to attract incorruptible moral officials?
  - Under what conditions can governments bring them in (what sort of incentives?)
  - Topic of Dal Bó et al (2013) in two weeks

Sanchez de la Sierra (2020): On the Origins of the State: Bandits and Taxation in Eastern Congo [Student Presentation]

Takeaways and remaining questions

Capable states are the foundation for economic and institutional development

- Able to collect taxes (or any resources) for common interests
- Provision of essential public services
- Protect market activities with legal measures
- Role played by how much significance of common interest and civic movements
  - Internal conflicts debilitate state capacity, external conflicts likely raise it
- Weak states experience corruption and state capture (leading to further corruption)

How are we doing this empirically

- How do developing countries create policy instruments to enhance state capacity?
  - Field and natural experiments on fiscal capacity
- Studies on the bureaucrats implementing state operations
  - How to motivate and monitor them?